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1 May 1974
USED IN LIEU OF
MIL-M-38510/41(USAF)
31 May 1973
SUPERSEDING
MIL-M-0038510/41A(USAF)
15 October 1973

MILITARY SPECIFICATION

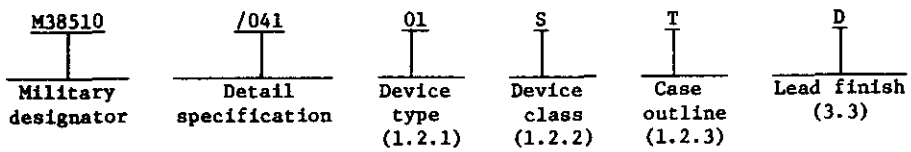
MICROCIRCUITS, DIGITAL, TTL, LOW POWER AND-OR-INVERT GATES, MONOLITHIC SILICON

This limited coordination Military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/41(USAF). It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, TTL, low power, AND-OR-INVERT logic gating microcircuits for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:



1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit	Similar commercial type
01	Dual, 2-wide, AND-OR-INVERT gate	54L51
04	4-wide, 3-2-2-3 input AND-OR-INVERT gate	54L54

1.2.2 Device class. The device class shall be Class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

Supply voltage range	0 Vdc to 8.0 Vdc
Input voltage range	0 Vdc to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Lead temperature (soldering, 10 seconds)	300°C

1.4 Recommended operating conditions.

Ambient operating temperature range	-55°C to 125°C
Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.7 Vdc

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2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight and physical dimensions. The design, construction, weight and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C and 125°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510).

3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figure 4 shall be performed at 25°C. All devices shall meet the limits specified in table II.

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TABLE I. (Device type 01) Electrical characteristics tests.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
1*	V_{OL}	2.0 V	GND	GND	4.5 V	GND						GND	2.0 mA	2.0 V	2.0 V		0.3	±0.03	V
2*			2.0 V	2.0 V		2.0 V							2.0 mA	GND	GND				
3*							2.0 V	2.0 V	GND	GND	2.0 mA								
4*	V_{OL}						GND	GND	2.0 V	2.0 V	2.0 mA						0.3	±0.03	
5*	V_{OH}	0.7 V	0.7 V	5.5 V		5.5 V							-100 µA	5.5 V	5.5 V	2.4		±0.12	
6*			0.7 V	5.5 V		5.5 V							-100 µA						
7*							0.7 V	5.5 V	5.5 V	5.5 V	-100 µA								
8*	V_{OH}				4.5 V		5.5 V	5.5 V	0.7 V	5.5 V	-100 µA					2.4		±0.12	V
9	I_{OS}	GND	GND	GND	5.5 V	GND							GND	GND	GND	-3	-15		mA
10	I_{OS}						GND	GND	GND	GND	GND					-3	-15		mA
11*	I_{IH1}	2.4 V	GND	GND		GND								GND	GND		10	10X or 1 µA	µA
12*		GND	2.4 V	GND		GND													
13*			GND	2.4 V		GND													
14*				GND		2.4 V	GND												
15*						GND	2.4 V	GND											
16*							GND	2.4 V	GND										
17*								GND	2.4 V	GND				GND					
18*									GND	2.4 V				GND					
19*										GND				2.4 V	GND				
20*	I_{IH1}	GND												GND	2.4 V		10	10X or 1 µA	
21	I_{IH2}	5.5 V	GND												GND		100		
22		GND	5.5 V	GND															
23			GND	5.5 V		GND													
24				GND		5.5 V	GND												
25						GND	5.5 V	GND											
26							GND	5.5 V	GND										
27								GND	5.5 V	GND									
28								GND	GND	5.5 V				GND					
29	I_{IH2}	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND		GND		5.5 V	GND		100		µA

See notes at end of table.

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TABLE I. (Device type 01) Electrical characteristics tests.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
30	I_{IH2}	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND		GND		GND	5.5 V		100		µA
31	I_{IL}	0.3 V	5.5 V	5.5 V	↓	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		↓		5.5 V	↓		-.18		mA
32	↓	5.5 V	0.3 V	5.5 V		5.5 V	↓	↓	↓	↓				↓			↓		↓
33		↓	5.5 V	0.3 V		5.5 V	↓	↓	↓	↓				↓					
34			↓	5.5 V		0.3 V	5.5 V	↓	↓	↓				↓					
35				↓		5.5 V	0.3 V	5.5 V	↓	↓				↓					
36						↓	5.5 V	0.3 V	5.5 V	↓				↓					
37							↓	5.5 V	0.3 V	5.5 V				↓					
38								↓	5.5 V	0.3 V				5.5 V	↓				
39	↓	↓	↓	↓		↓	↓	↓	5.5 V	5.5 V				0.3 V	5.5 V		↓		
40	I_{IL}	5.5 V	5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				5.5 V	0.3 V		-.18		
41	I_{CCL}	5.0 V	5.0 V	5.0 V	↓	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V		↓		5.0 V	5.0 V		1.3		
42	I_{CCH}	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND		GND		GND	GND		0.8		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurements"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE I. (Device type 04) Electrical characteristics tests.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
1*	V_{OL}	2.0 V	GND	GND	4.5 V	GND	GND	GND	GND	GND	2.0 mA	GND		2.0 V	2.0 V		0.3	±0.03	V
2*		GND	2.0 V	2.0 V		GND	GND	GND	GND	GND				GND	GND				
3*		GND	GND	GND		2.0 V	2.0 V	GND	GND	GND				GND	GND				
4*	V_{OL}	GND	GND	GND		GND	GND	2.0 V	2.0 V	2.0 V	2.0 mA			GND	GND		0.3	±0.03	
5*	V_{OH}	0.7 V	0.7 V	5.5 V		0.7 V	5.5 V	5.5 V	0.7 V	5.5 V	-100 µA			0.7 V	5.5 V	2.4		±0.12	
6*																			
7*																			
8*	V_{OH}	0.7 V	0.7 V	5.5 V	4.5 V	0.7 V	5.5 V	5.5 V	0.7 V	5.5 V	-100 µA			0.7 V	5.5 V	2.4		±0.12	V
9	I_{OS}	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND	GND			GND	GND	-3	-15		mA
10*	I_{IH1}	2.4 V	GND	GND		GND											10	10X or 1 µA	µA
11*		GND	2.4 V	GND		GND													
12*			GND	2.4 V		GND													
13*				GND		2.4 V	GND												
14*						GND	2.4 V	GND											
15*							GND	2.4 V	GND										
16*								GND	2.4 V	GND									
17*									GND	2.4 V				GND					
18*										GND				2.4 V	GND				
19*	I_{IH1}	GND												GND	2.4 V		10	10X or 1 µA	
20	I_{IH2}	5.5 V	GND												GND		100		
21		GND	5.5 V	GND															
22			GND	5.5 V		GND													
23				GND		5.5 V	GND												
24						GND	5.5 V	GND											
25							GND	5.5 V	GND										
26								GND	5.5 V	GND									
27									GND	5.5 V				GND					
28									GND	GND				5.5 V	GND				
29	I_{IH2}	GND	GND	GND		GND	GND	GND	GND	GND				GND	5.5 V		100		µA
30	I_{IL}	0.3 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		GND		5.5 V	5.5 V		-.18		mA

See notes at end of table.

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TABLE I. (Device type 04) Electrical characteristics tests.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
31	I_{IL}	5.5 V	0.3 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		GND		5.5 V	5.5 V		-.18		mA
32	↓	↓	5.5 V	0.3 V	↓	5.5 V	5.5 V	↓	↓	↓		↓		↓	↓		↓		↓
33			↓	5.5 V		0.3 V	5.5 V	↓											
34				↓		5.5 V	0.3 V	5.5 V	↓										
35						↓	5.5 V	0.3 V	5.5 V	↓									
36							↓	5.5 V	0.3 V	5.5 V				↓					
37								↓	5.5 V	0.3 V				5.5 V	↓				
38	↓	↓	↓	↓		↓	↓	↓	5.5 V	5.5 V				0.3 V	5.5 V		↓		
39	I_{IL}	5.5 V	5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				5.5 V	0.3 V		-.18		
40	I_{CCL}	5.0 V	5.0 V	5.0 V	↓	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V		↓		5.0 V	5.0 V		0.99		↓
41	I_{CCH}	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND		GND		GND	GND		0.8		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurements"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

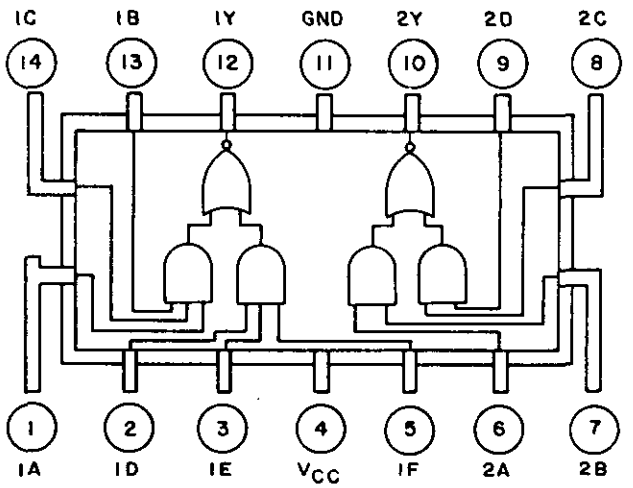
TABLE II. (Device types 01 and 04) Switching characteristics.

Characteristic	Symbol	Conditions 1/	Test figure	Temp.	Limits Min	Limits Max	Units
Propagation delay time, high-to-low level	t_{PHL}	V_{CC} = 5.0 V C_L = 50 pF R_L = 4KΩ ±5%	4	25°C 125°C -55°C	3 3 3	66 99 99	ns
Propagation delay time, low-to-high level	t_{PLH}			25°C 125°C -55°C	3 3 3	90 120 120	ns

1/ Check each gate.

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Device type 01



Device type 04

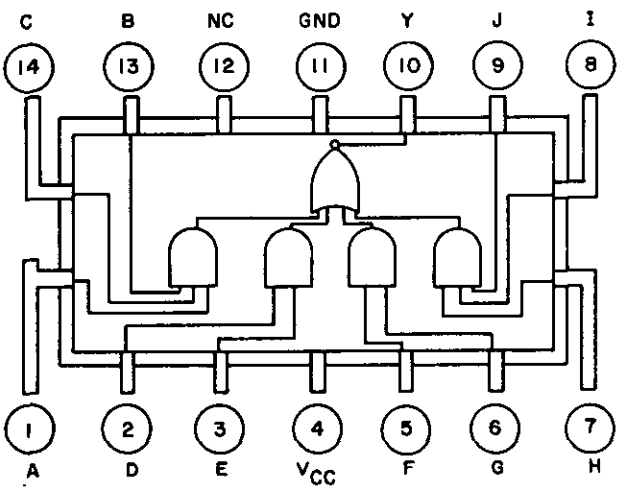


FIGURE 1. Logic diagram and terminal connections.

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DEVICE TYPE 01, GATE 1

INPUTS						OUTPUT
1A	1B	1C	1D	1E	1F	1Y
H	H	H	*	*	*	L
*	*	*	H	H	H	L

DEVICE TYPE 01, GATE 2

INPUTS				OUTPUT
2A	2B	2C	2D	2Y
H	H	*	*	L
*	*	H	H	L

*Either H or L.
All other combinations of an L on one input of each AND gate and either H or L on the remaining inputs give H output.

Positive Logic: $1Y = \overline{(1A \cdot 1B \cdot 1C) + (1D \cdot 1E \cdot 1F)}$

$2Y = \overline{(2A \cdot 2B) + (2C \cdot 2D)}$

DEVICE TYPE 04

INPUTS										OUTPUT
A	B	C	D	E	F	G	H	I	J	Y
H	H	H	*	*	*	*	*	*	*	L
*	*	*	H	H	*	*	*	*	*	L
*	*	*	*	*	H	H	*	*	*	L
*	*	*	*	*	*	*	H	H	H	L

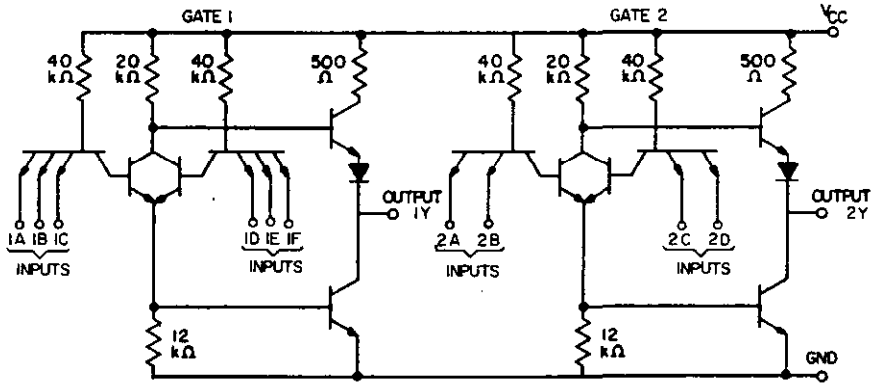
*Either H or L.
All other combinations of an L on one input of each AND gate and either H or L on the remaining inputs give H output.

Positive Logic: $Y = \overline{(A \cdot B \cdot C) + (D \cdot E) + (F \cdot G) + (H \cdot I \cdot J)}$

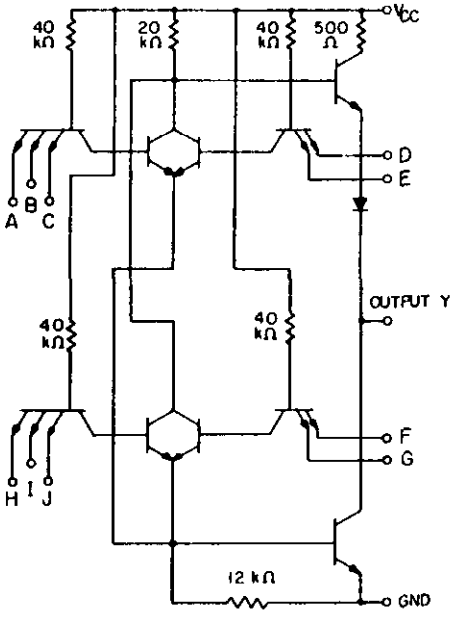
FIGURE 2. Truth tables and logic equations.

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Device type 01



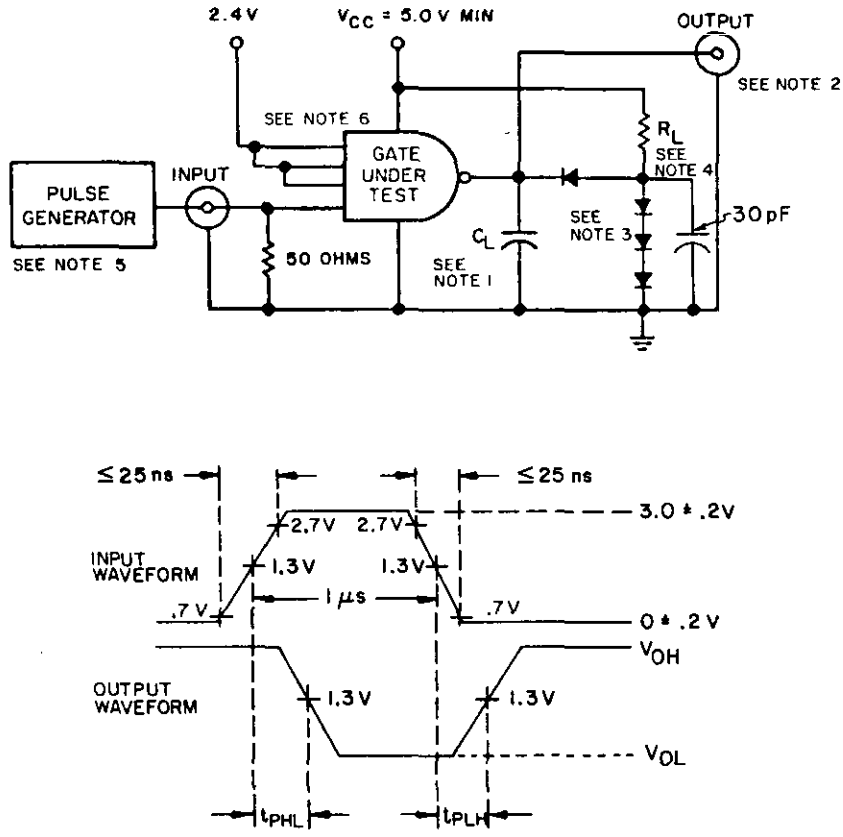
Device type 04



NOTE: Component values shown are nominal.

FIGURE 3. Schematic circuits.

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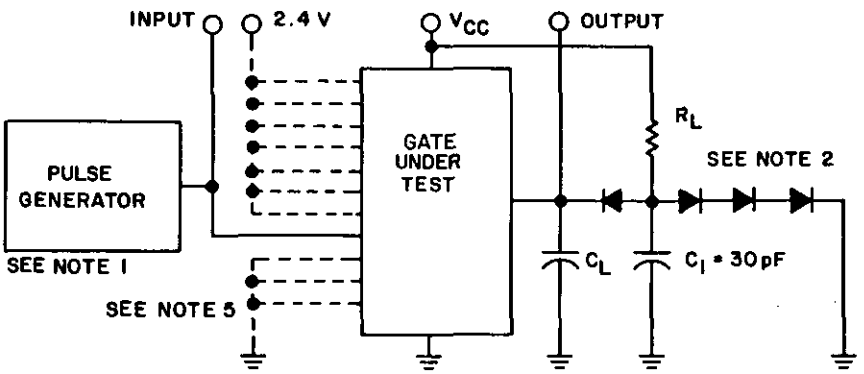


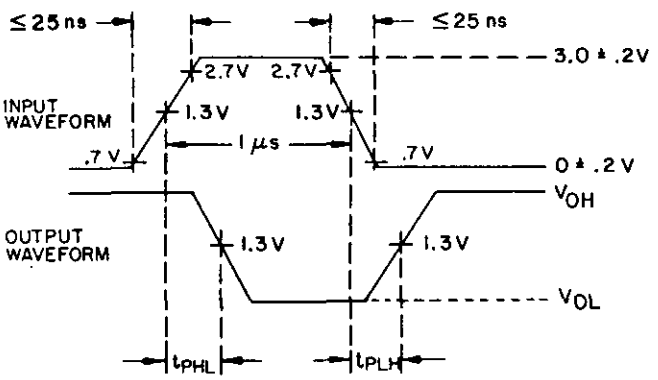
Device type 01

NOTES:
1. C_L = 50 pF, including scope probe, wiring, and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.
4. R_L = 4000 ohms ±5%.
5. Pulse generator Z_{OUT} ≈ 50 ohms, PRR = 1 MHz.
6. Connect all inputs of unused AND sections to ground.

FIGURE 4. Switching time test circuit and waveform.

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Device type 04

NOTES:
1. Pulse generator $Z_{out} \approx 50$ ohms, PRR $\leq$ 500 kHz.
2. All diodes are 1N916 or equivalent.
3. $t_{PD} = \frac{t_{PHL} + t_{PLH}}{2}$
4. C_L = 50pF minimum including scope probe, wiring, and stray capacitance, without package in test fixture.
5. Connect input pulse to one input of an AND section and 2.4V to all inputs of that AND section. All other unused AND sections are grounded.

FIGURE 4. Switching time test circuit and waveform - Continued.

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3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Product assurance provisions. Product assurance shall be in accordance with MIL-M-0038510.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be as specified in MIL-M-0038510.

6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-31)

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